IEEE GUB Student Branch

Green University of Bangladesh

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Header

The IEEE GUB Student Branch, Green University of Bangladesh, organised a workshop on Software Quality Assurance (SQA), co-organised by IEEE Computer Society (IEEE CS), IEEE Power and Energy Society (IEEE PES), and IEEE Women in Engineering (IEEE WIE). The workshop introduced participants to software testing fundamentals, manual and automation testing, security testing, test case design, bug reporting, and software development methodologies. It also highlighted the role of AI in modern software quality assurance, providing students with practical insights and industry-oriented guidance to prepare for careers in the field.

Description

The workshop began with an introduction to Software Quality Assurance (SQA), outlining its role in ensuring software quality and reliability throughout the development process. The session introduced three major areas of testing: manual testing, automation testing, and security testing. Participants were given an introductory overview of the SQA career roadmap, with emphasis on progressing from fundamental knowledge to intermediate-level proficiency before advancing to more specialised skills. The discussion highlighted that while beginners should start with the basics, entering the software industry requires a deeper understanding of testing concepts, practical application, and industry expectations.

A key part of the workshop focused on the importance of identifying and resolving defects during the development phase rather than allowing them to reach production. The speaker discussed the potential impact of software failures on businesses, users, and critical systems, emphasising the value of early testing and preventive quality practices. The session also introduced the Waterfall and Agile software development models, explaining their relevance to development workflows and the integration of testing into different stages of the software development lifecycle.

To demonstrate why software quality matters, the workshop presented three widely remembered software failures: the Ariane 5 rocket failure in 1996, the Knight Capital trading incident in 2012, and the CrowdStrike update incident in 2024. These examples illustrated how software defects, inadequate validation, or problematic updates can lead to significant financial losses, operational disruption, and serious consequences. The discussion reinforced the idea that software quality is not merely a final checkpoint but an essential responsibility throughout development and deployment.

The session also explored how the responsibilities of Quality Assurance (QA) engineers are evolving with the increasing use of AI-driven development tools and AI-assisted coding practices. As more developers and users rely on AI to generate code, the need to evaluate outputs, identify defects, validate functionality, and assess reliability becomes increasingly important. The speaker highlighted the growing relevance of critical evaluation and systematic testing, particularly when AI-generated code is used without sufficient understanding of its limitations. This discussion introduced participants to the changing expectations of QA professionals in an increasingly AI-driven software industry.

Another segment focused on the career progression from software tester to quality leader. The speaker discussed the competitive nature of the job market and emphasised the importance of preparing early for industry opportunities. Rather than relying solely on specific testing frameworks or tools, participants were encouraged to strengthen their fundamentals, develop process-oriented thinking, and learn to approach problems in a structured and organised manner. The discussion also touched on salary expectations, market demand, and the potential growth of specialised QA responsibilities as software systems and AI-based applications become more complex.

The workshop introduced the habits and mindset of a professional QA engineer through a practical bottle-testing example. By contrasting a normal mindset with a quality-focused mindset, the session demonstrated how effective testing requires more than checking whether a product works under ordinary conditions. A quality-focused approach involves exploring different scenarios, identifying possible failure points, questioning assumptions, and considering how a product might behave under unexpected conditions. This example encouraged participants to think critically and approach testing from the perspective of reliability, usability, and user expectations.

The speaker then provided a brief overview of the Software Development Life Cycle (SDLC), explaining the major stages involved in developing and maintaining software. The discussion connected software testing with the broader development process and highlighted the importance of collaboration among

developers, testers, and other technical professionals. DevOps was also introduced as an approach that connects development and operations through collaboration, automation, and continuous delivery practices. The session noted that opportunities to work with mature DevOps practices may vary across the local technology industry, making it important for aspiring professionals to understand the underlying concepts and gradually build relevant skills.

The workshop subsequently covered the fundamentals of software testing, including the practical roles of tools such as Jira and Microsoft Excel. Participants were introduced to how these tools can support test management, issue tracking, documentation, and the organisation of testing activities. The session also discussed different levels and types of software testing, helping participants understand that testing can be performed at different stages and for different purposes depending on the requirements of a software system.

A dedicated segment focused on the anatomy of a test case and the essential information required to create effective test cases. The speaker emphasised the importance of documenting test cases in a structured tabular format, with clearly defined fields to support consistency, traceability, and repeatability. The discussion highlighted the need to specify relevant test information, define expected outcomes, and organise test scenarios so that software behaviour can be evaluated systematically. This guidance helped establish the importance of documentation as a core component of professional testing practice.

The workshop also explained the principles of effective bug reporting. Participants learned about the information required to communicate a defect clearly, including reproducible steps, the observed behaviour, and the expected result. The discussion introduced the concepts of severity and priority, clarifying the distinction between the impact of a defect and the urgency with which it should be addressed. The speaker further outlined a systematic bug-hunting approach, encouraging participants to investigate software behaviour methodically and provide developers with clear, actionable information that supports efficient defect resolution.

Another important topic was test automation and the principle of using code to test code. The session explained how automation can reduce repetitive manual work, improve consistency, and support more efficient testing workflows. Participants were introduced to the role of Git and GitHub in version control and collaborative software development, alongside an overview of Continuous Integration and Continuous Delivery or Deployment (CI/CD). The speaker stressed that the choice of automation framework is less important than understanding testing fundamentals, writing meaningful tests, and applying automation to solve practical problems. The discussion also highlighted the essential skills required for both manual and automation testing, encouraging participants to build a strong foundation before specialising in particular tools.

The session further emphasised the importance of understanding programming languages and developing an architectural perspective when preparing for technical roles. Rather than focusing exclusively on syntax or individual frameworks, aspiring QA professionals were encouraged to understand how software components interact, how systems are structured, and how defects can emerge across different layers of an application. The workshop concluded its career-focused discussion by addressing six important skill gaps commonly observed by recruiters in fresh graduates, reinforcing the need for continuous learning, practical experience, analytical thinking, and effective communication when preparing for industry roles.

Overall, the workshop provided participants with a broad introduction to software quality assurance, from fundamental testing concepts and documentation practices to automation, CI/CD, AI-assisted development, and long-term career preparation. It encouraged students to approach software quality as a continuous process, strengthen their technical fundamentals, and develop the organised thinking required to identify defects and build more reliable software.

Footer

The workshop introduced participants to Software Quality Assurance, covering testing methodologies, test case design, bug reporting, automation, and the growing role of AI in software development. It encouraged students to strengthen their fundamentals, develop practical skills, and prepare for industry opportunities. Through this initiative, the IEEE GUB Student Branch reaffirmed its commitment to technical education and professional development.

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